

Relational Algebra in DBMS

What is Relational Algebra?

- Relational Algebra is a procedural query language used in Database Management Systems.
- It performs operations on tables (relations).
- It forms the theoretical foundation for SQL.

Definition

Relational Algebra:

A collection of operations used to retrieve and manipulate data from relational databases.

Important Features

1. Operates on relations (tables)
2. Produces new relation as output
3. Procedural language
4. Used internally by DBMS
5. Basis for SQL queries

Basic Terminology

Term	Meaning
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Relation	Table
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Tuple	Row
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Attribute	Column
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Example Tables

STUDENT Table

SID Name Dept

101 Ravi CSE

102 Anu ECE

103 Kiran CSE

MARKS Table

SID Marks

101 85

102 90

103 75

Main Operations in Relational Algebra

1. Select (σ)
 2. Project (π)
 3. Union ($\cup$)
 4. Set Difference ($-$)
 5. Cartesian Product ($\times$)
 6. Rename (ρ)
 7. Intersection ($\cap$)
 8. Join ($\bowtie$)
 9. Division ($\div$)
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1. SELECT Operation (σ)

Purpose

- Selects rows that satisfy a condition.
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Syntax

σ condition (Relation)

Example

Find students from CSE department.

σ Dept='CSE' (STUDENT)

Step-by-Step Explanation

Step 1

Check each row.

Step 2

Select rows where Dept = CSE.

Output

SID	Name	Dept
101	Ravi	CSE
103	Kiran	CSE

Important Points

6. Select works on rows
7. Uses conditions
8. Similar to WHERE clause in SQL

2. PROJECT Operation (π)

Purpose

- Selects specific columns.

Syntax

π attribute_name (Relation)

Example

Display only Name column.

π Name (STUDENT)

Step-by-Step Explanation

Step 1

Take STUDENT table.

Step 2

Display only Name column.

Output

Name

Ravi

Anu

Kiran

Important Points

9. Project works on columns
10. Duplicate rows removed automatically

3. UNION Operation (U)

Purpose

- Combines rows from two relations.
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Condition

- Tables must have same number of columns and compatible datatypes.
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Syntax

Relation1 U Relation2

Example

Combine two student lists.

Important Points

11. Removes duplicate rows
12. Similar structure required

4. SET DIFFERENCE Operation (-)

Purpose

- Displays rows present in first table but not in second.

Syntax

Relation1 - Relation2

Example

Students in CSE but not in ECE.

Important Points

- 13. Finds unique rows
- 14. Tables must be compatible

5. CARTESIAN PRODUCT ($\times$)

Purpose

- Combines every row of first table with every row of second table.

Syntax

Relation1 $\times$ Relation2

Step-by-Step Example

Suppose:

Table A

A

1

2

Table B

B

X

Y

Result

A B

1 X

1 Y

A B

2 X

2 Y

Important Points

15. Produces large output

16. Basis for joins

6. RENAME Operation (ρ)

Purpose

- Renames relation or attributes.

Syntax

ρ NewName (Relation)

Example

ρ STUD (STUDENT)

Important Points

17. Improves readability

18. Useful in joins

7. INTERSECTION Operation ($\cap$)

Purpose

- Displays common rows in both tables.

Syntax

Relation1 $\cap$ Relation2

Important Points

- 19. Shows shared records
- 20. Tables must be compatible

8. JOIN Operation ($\bowtie$)

Purpose

- Combines related rows from two tables.

Syntax

Relation1 $\bowtie$ condition Relation2

Example

Join STUDENT and MARKS tables.

STUDENT $\bowtie$ STUDENT.SID = MARKS.SID MARKS

Step-by-Step Explanation

Step 1

Compare SID values.

Step 2

Combine matching rows.

Output

SID Name Dept Marks

101 Ravi CSE 85

SID Name Dept Marks

102 Anu ECE 90

103 Kiran CSE 75

Important Points

21. Most important operation

22. Combines related data

Types of Joins

23. Theta Join

24. Equi Join

25. Natural Join

9. DIVISION Operation ($\div$)**Purpose**

- Finds values related to all values in another relation.

Example

Students who completed all subjects.

Important Points

26. Advanced operation

27. Used less frequently

Compound Expressions**Example**

Find names of CSE students.

π Name (σ Dept='CSE' (STUDENT))

Step-by-Step Explanation

Step 1

Select CSE students.

Step 2

Project Name column.

Relational Algebra Symbols

Symbol Operation

σ	Select
π	Project
$\cup$	Union
$-$	Difference
$\times$	Cartesian Product
$\bowtie$	Join
$\cap$	Intersection
ρ	Rename
$\div$	Division

Advantages of Relational Algebra

- 28.Simple mathematical foundation
- 29.Basis for SQL
- 30.Helps query optimization
- 31.Easy data manipulation
- 32.Supports relational model
- 33.Improves understanding

34.Used internally by DBMS

35.Provides procedural approach

Disadvantages

36.Difficult for beginners

37.Uses mathematical symbols

38.Not user-friendly like SQL

Real-Time Example

College Database

Query

Find names of students in CSE department.

Relational Algebra

π Name (σ Dept='CSE' (STUDENT))

Difference Between SQL and Relational Algebra

SQL

Declarative language

User-friendly

Used by users

Relational Algebra

Procedural language

Mathematical

Used internally

Additional Important Points

39.Relational algebra works on tables.

40.Output is always a relation.

41.Select filters rows.

42.Project filters columns.

43.Join combines tables.

- 44. Cartesian product creates combinations.
- 45. Union combines relations.
- 46. Intersection finds common rows.
- 47. Rename changes relation name.
- 48. Division is advanced operation.
- 49. Relational algebra is important for DBMS theory.
- 50. SQL is based on relational algebra.

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